

# Outage Investigation Report —

## OIR-2025-001

---

| Field             | Value                                                                           |
|-------------------|---------------------------------------------------------------------------------|
| Document ID       | OIR-2025-001                                                                    |
| Report Date       | 2025-04-28                                                                      |
| Author            | Marcus Chen, Sr. Reliability Engineer                                           |
| Reviewing Manager | Priya Ramachandran, Plant Manager, NW-PowerPool                                 |
| Region            | NW-PowerPool                                                                    |
| Site              | NW-PowerPool-WIND-O24                                                           |
| Asset             | SUB-10629 (Hitachi Energy, substation_equipment)                                |
| Outage ID         | OUT-2025-1601                                                                   |
| Classification    | Forced outage — soc_drift                                                       |
| Distribution      | NW-PowerPool Operations, Reliability Engineering, Compliance, Vendor Management |

---

## 1. Outage Summary

A forced outage was declared at NW-PowerPool-WIND-O24 on the SUB-10629 substation interconnect on 2025-04-11 at 17:43 PDT. The asset remained offline until 2025-04-19 at 20:43 PDT, for a duration of **195 hours**. Lost generation during the event was **4,754.8 MWh**, the largest single-asset NW-PowerPool event of Q2 2025. Estimated revenue impact: \$285,288.

The event was triggered by a state-of-charge (SoC) drift alarm on the co-located battery buffer feeding the 34.5 kV collector. SoC drift propagated to the substation protection relays, which tripped the interconnect on under-voltage. The asset has now suffered 4 forced outages in FY2025 (refer to repeat-offender summary appended to RCA-2025-002).

## 2. Timeline of Events

| Timestamp (PDT)         | Event                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-04-11 14:02        | SCADA flags SoC drift >2.3% on collector battery buffer Bank B-3. Alarm classified "advisory" per CSM-7 §4.2.                                                |
| 2025-04-11 16:11        | Drift widens to 3.9%. Operator on shift (T. Okafor) acknowledges alarm; no escalation per current advisory thresholds.                                       |
| 2025-04-11 17:43        | Protection relay R-12B trips on under-voltage. SUB-10629 isolates from the 230 kV bus. Asset declared forced outage.                                         |
| 2025-04-11 18:02        | Control room contacts Cascadia Power Services (CPS) on-call. CPS estimates 6-hour mobilization to NW-PowerPool-WIND-024 due to weather (Cascade Pass icing). |
| 2025-04-11 19:30        | Mobilization estimate revised to 11 hours. Per EPA-2024-11, on-call manager invoked Emergency Procurement Authorization.                                     |
| 2025-04-11 21:45        | Apex Emergency Grid Solutions accepts dispatch. Crew arrives onsite 2025-04-12 04:10.                                                                        |
| 2025-04-12 06:00        | Apex confirms relay R-12B operated correctly. Root issue suspected at battery management system (BMS) firmware on Bank B-3.                                  |
| 2025-04-13 – 2025-04-17 | BMS replacement, relay re-coordination, and post-energization tests. Two failed energization attempts on 2025-04-15 due to residual harmonics.               |
| 2025-04-19 20:43        | Asset returned to service.                                                                                                                                   |

## 3. Telemetry Observations

- SoC drift trended from baseline 0.4% (rolling 30-day) to 3.9% over 3 hours and 41 minutes prior to trip. Threshold for "action required" under Standard MS-2025-03 is 2.0% sustained > 30 min.
- Bus voltage on the 34.5 kV side dropped from 34.2 kV to 31.8 kV in the final 90 seconds before the trip.
- Bank B-3 cell temperature spread widened from 4.1 °C to 11.7 °C between adjacent modules, indicating BMS balancing failure.
- Ambient temperature 6 °C; gusting NW wind 19–24 m/s. Conditions within seasonal norm.

- The same drift signature was logged on Bank B-3 on 2025-02-19 and 2025-03-04 without trip — the WO-2025-04118 inspection at the end of February was closed as "PM completed, no anomalies" by Pinnacle Energy Maintenance. See documentation conflict noted in CAR-2025-004.

## 4. Restoration Activities

| Vendor                                 | Scope                                                                                                    | Hours                                           | Invoice (USD)           |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------|
| Apex<br>Emergency Grid<br>Solutions    | Onsite diagnostics, BMS module replacement (4 modules), relay re-coordination, two energization attempts | 187 (incl. 41 weekend hours at 1.5× multiplier) | \$412,000 (preliminary) |
| Hitachi Energy<br>(OEM remote support) | Firmware reconciliation, factory acceptance test sign-off                                                | 14                                              | \$38,500                |
| Total                                  |                                                                                                          |                                                 | <b>\$450,500</b>        |

For comparison, CPS estimate for the equivalent scheduled scope (provided 2025-04-22 retrospective) was \$98,600. Reference EMR-2025-001 for emergency invoicing detail.

## 5. Preliminary Root Cause

Battery Management System firmware on Bank B-3 was running version 2.7.1, scheduled for upgrade to 2.7.4 under the Q1 2025 preventive maintenance plan. The upgrade WO (WO-2025-02041) was one of the 14 preventive items deferred under the Q1 2025 budget freeze (see CAR-2024-118 and CAR-2025-002). Version 2.7.1 has a known cell-balancing defect under cold-soak conditions documented in Hitachi Energy Service Bulletin SB-HE-2024-09.

The immediate technical cause was BMS firmware defect. The procedural cause was deferral of a known-corrective firmware update.

## 6. Recommendations

1. Expedite remaining Q1 2025 deferred PM work orders at NW-PowerPool-WIND-024. Track under CAR-2025-002.
2. Reclassify SoC drift > 2.0% as an immediate action threshold under MS-2025-03, harmonized with CSM-7.

3. Audit Pinnacle Energy Maintenance close-out documentation for WO-2025-04118 and related inspections at NW-PowerPool-WIND-024. Cross-reference CAR-2025-004 (Pinnacle documentation drift).
4. Add SUB-10629 to the watchlist for asset-level replacement planning. See RAM-2025-002 in preparation.
5. Cross-reference RCA-2025-002 (Fall Overheating Sequence) which traces an additional SUB-10629 event on 2025-11-12 to a related deferred PM thread.

## Signatures

| Role              | Name                                  | Date       |
|-------------------|---------------------------------------|------------|
| Author            | Marcus Chen, Sr. Reliability Engineer | 2025-04-28 |
| Plant Manager     | Priya Ramachandran                    | 2025-04-29 |
| Compliance Review | D. Whitcombe, NERC Compliance Lead    | 2025-04-30 |

**Distribution:** NW-PowerPool Operations · Reliability Engineering · Compliance · Vendor Management · CFO Office (cost flag)

**References:** CAR-2024-118 · CAR-2025-002 · CAR-2025-004 · EMR-2025-001 · RCA-2025-002 · RAM-2025-002 (draft) · Hitachi Service Bulletin SB-HE-2024-09 · Standard MS-2025-03 · CSM-7